VOCALSHIELD AI - ENTERPRISE DEFENSE PLATFORM

Real-Time Voice Cloning Detection & Enterprise Impersonation Defense | SIH 2026 PS-26104

1. EXECUTIVE OVERVIEW

VocalShield AI is an enterprise-grade Security Operations Center (SOC) defense system engineered to detect and neutralize synthetic speech deepfakes, AI voice clones, and executive impersonation attacks in real time across enterprise VoIP, PBX, and banking telephony channels.

Operating with sub-45ms DSP latency, VocalShield decouples Voice Authenticity (Is the speech synthetic or biological?) from Speaker Identity (Does it match the claimed authorized executive?), defeating both AI Voice Clones and Human Impostors before fraudulent financial authorizations can execute.

" Problem Statement: Smart India Hackathon (SIH 2026) | PS-26104
" Domain: Telephony Cybersecurity, Core Banking & Treasury Fraud Defense
" Target Channels: Enterprise VoIP, SIP PBX, Contact Centers, Mobile Banking Trunks
" Compliance: DPDP Act 2023 | CERT-In Guidelines | Zero Raw Audio Retention

2. CORE DEFENSE ENGINE & MULTI-SIGNAL MATHEMATICAL FUSION

7-Signal Mathematical & Acoustic Fusion Matrix

VocalShield evaluates 7 independent mathematical and acoustic signals to produce a unified, tamper-evident Risk Index (0 - 100):

1. Synthetic Speech Probability: Detects phase smearing, non-linear distortion, and mathematical artifacts characteristic of modern neural vocoders (HiFi-GAN, WaveGlow,
" Diffusion).
2. Speaker Identity Biometric Match: Computes cosine similarity between real-time
" 192-dimensional d-vectors and pre-enrolled authorized executive voiceprints.
3. Neural Vocoder Artifact Detection: Evaluates high-frequency spectral
" discontinuities and phase incoherence in the >7.8 kHz acoustic band.
4. Spectral Anomaly Analysis: Identifies unnatural brickwall roll-offs, frequency
" truncation, and spectral voids produced by generative AI audio pipelines.
5. Prosody Consistency & Pitch Micro-Tremor: Measures F0 glottal jitter. Biological human vocal folds exhibit natural micro-tremor (0.5% - 1.5%), whereas synthetic TTS
" models display robotic pitch stability (<0.2%).
6. Loudspeaker Replay & Transducer Analysis: Evaluates Total Harmonic Distortion (THD) and room impulse response echo tails to identify acoustic presentation attacks
" (replaying audio through speakers).
7. Biological Voice Authenticity Score: Assesses natural vocal tract formants (F1-F4)
" and organic breathing pause rhythms unique to biological humans.

3. DUAL-VERIFICATION MATRIX & ARCHITECTURE

Traditional voice biometrics fail against AI clones because a high-quality clone matches the speaker's vocal characteristics. VocalShield separates Authenticity from Identity:

Legitimate Executive: Genuine Authenticity (98%) + Matched Identity (96%) -> Allow /
" Green
Human Impostor Attack: Genuine Authenticity (97%) + Mismatched Identity (12%) ->

" Alert & Freeze
AI Executive Clone Attack: Fake Authenticity (94%) + Matched Identity (92%) ->
" Critical Interlock & Banking Freeze
Automated Voicebot Attack: Fake Authenticity (99%) + Mismatched Identity (8%) ->
" Immediate Trunk Drop

4. REAL-TIME SOC & TELEPHONY SURVEILLANCE MODULES

Executive SOC Dashboard & Live Threat Radar: Situational awareness dashboard with active call counters, threat distribution charts, real-time risk gauges, and
" prioritized alert feeds (Verified, Suspicious, High Risk, Critical Spoof).
Real-Time Telephony Call Monitor: Active call surveillance designed for SIP/VoIP enterprise PBX networks with caller ID spoofing detection (detecting mismatched
" carrier metadata vs. caller).
Live Call HUD: Heads-Up Display featuring real-time audio waveforms, dynamic risk
" gauges, and synchronized transcription with flagged high-risk keywords.
Continuous In-Call Voice Authentication: Rolling-window biometric verification
" throughout ongoing calls to detect mid-call speaker switching or voice hijacking.

5. AUDIO DSP LAB & DEEP FORENSICS

Live Microphone Lab: In-browser Web Audio API DSP probe with sub-45ms latency,
" 256-bin FFT spectrum visualizer, oscilloscope, and SNR metering.
Deep Audio Forensics File Upload: Drag-and-drop batch or single file analysis (.wav,
" .mp3, .m4a, .flac) with synchronized timeline playback scrubber.
Granular Parameter Inspection: Detailed breakdown of jitter, shimmer, formant
" stability, spectral centroid, and transducer anomalies.

6. CROSS-CALL THREAT INTELLIGENCE & ATTACKER CLUSTERING

Acoustic Centroid Clustering: Matches voice attack signatures across multiple
" unrelated phone numbers and spoofed caller IDs using acoustic embedding distances.
Syndicate & Campaign Tracking: Identifies coordinated syndicate operations targeting
" corporate treasuries across multiple branches or departments.
Attacker Profile Dossiers: Aggregates historical attack attempts, associated phone
" numbers, targeted executives, and vocoder signatures.

7. AUTOMATED DEFENSE INTERLOCKS & ADAPTIVE POLICIES

Core Banking Freeze: Automatically halts RTGS, NEFT, and SWIFT transaction approvals
" when a high-confidence voice clone is detected.
Out-of-Band (OOB) PSTN Callback: Enforces secondary telephone/PSTN verification prior
" to high-value transaction releases.
SIP Quarantine: Drops or isolates compromised telephony trunks from reaching
" executive extensions.
Dynamic Risk Thresholds: Stricter policies automatically applied based on financial
" transaction amounts (>INR 10 Lakhs) and threat tiers.

8. BIOMETRIC DIRECTORY & INDIAN LANGUAGE PROSODY NORMALIZATION

Trusted Voices Biometric Directory: Secure voiceprint enrollment for enterprise executives, CFOs, and treasury officers with cryptographic SHA-256 signatures and " drift tracking.

Indian Language Prosody Normalization: Calibrated across 8 Indian languages (Hindi, English, Tamil, Telugu, Kannada, Bengali, Marathi, Malayalam) to eliminate false positives caused by regional dialect cadences, syllable-timed phrasing, and native " intonations.

9. COMPLIANCE, CRYPTOGRAPHIC AUDIT LEDGER & AI REASONING

Tamper-Proof SHA-256 Audit Ledger: Chained block ledger architecture ensuring non-repudiation for legal and forensic evidence with built-in one-click integrity " verification.

Zero Raw Audio Retention: Audio buffers are discarded from RAM immediately after feature extraction; only cryptographic hashes are stored. Compliant with India’s DPDP " Act 2023.

Human-in-the-Loop Analyst Triage: Dedicated SOC Analyst Review Console for incident " management, case notes, ticket escalation, and false-positive calibration feedback.

Google Gemini 3.8 Flash Forensic Reasoning: Integrated LLM reasoning engine providing human-readable forensic audit reports explaining acoustic anomalies and recommended " mitigation steps in natural language.

10. SIH 2026 DEMO SCENARIOS & DEVELOPER SDK

" Scenario 1: Clean Human Call (Legitimate CFO high-value verification)

" Scenario 2: Diffusion Voice Clone (High-risk RTGS fraud attempt)

" Scenario 3: Robotic TTS Voicebot Attack (Automated social engineering bot)

" Scenario 4: Voice Conversion Impersonation (Attacker mimicking CEO pitch)

Enterprise Developer SDK & REST API Playground: Built-in API explorer with live endpoint testing for /api/v1/voice/verify, /api/v1/telephony/stream, and " /api/v1/ledger/audit with curl and Python code snippets.